MATH 4162/MATH 6201
Numerical methods for evolutionary differential equation
Fall 2025

COURSE PROFESSOR

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Access your course in `https://online.mun.ca/`

MEETING

The class is scheduled on TR from 09:00 to 10:15 in SN-3042.

OFFICE HOUR

Office hour: (T)R 1500-1600 in HH3054

Calendar description

This course covers numerical solution of initial value problems for ordinary differential equations by single and multi-step methods, Runge-Kutta, and predictor-corrector; numerical solution of boundary value problems for ordinary differential equations by shooting methods, finite differences and spectral methods; numerical solution of partial differential equations by the method of lines, finite differences, finite volumes and finite elements.

COURSE OUTLINE

We will start from stability of differential equations and approximation of derivatives. Then, we will move toward numerical methods covering finite difference methods, spectral methods, finite volume methods, and finite element methods. Additional topics arising from the discretization and solution of specific equations will be covered, i.e. stability, consistency, convergence, iterative methods, solvers by Fast Fourier Transforms (FFT), and implementation in python, matlab, and/or C++ programming.

Tentative lecture plan:

1. Discretization and stability of PDEs (2 weeks)
2. Finite Difference (FD) Methods (4 weeks)
3. Finite Volume (FV) Methods (2 weeks)
4. Spectral method (2 weeks)
5. Finite Element Methods (FEM) (2 weeks)

Learning Objectives

i) Traditional numerical solvers for modelling real world problems with examples drawing from energy and environment;

ii) Concept of modern data-driven numerical solvers for simulating complex systems;

iii) Utilization of data-science techniques for scientific analysis and presentation of modelling method.

Generative Artificial Intelligence (GAI)

The submission of work that has been created by generative artificial intelligence (GAI) tools and presented as a student's original work is considered an academic offence in this course, and your work will be subject to the appropriate Memorial's Academic Misconduct policy.

MARKING SCHEME

Assignments and case studies	25%		
Projects	30%		
Assignment 1		5%	Sep 26, 2025 at 1700
Case study 1		5%	Oct 10, 2025 at 1700
Midterm report		10%	Oct 17, 2025 at 1200
Assignment 2		5%	Oct 24, 2025 at 1700
Case study 2		5%	Nov 07, 2025 at 1700
Assignment 3		5%	Nov 21, 2025 at 1700
Final report		20%	Dec 04, 2025 at 2359
Final exam	45%		see exam schedule

PROJECT

The final report should have the format (and professional look) of a journal article. It should cover background of the problem chosen, discretization of governing equation and stability analysis, results and conclusion with future direction.

The project may be related to students' own research interests, but cannot be recycling of previous projects. The results do not have to be novel or original, but must be a sound investigation of an actual topic.

Students are strongly encouraged to consider a simulation or digital twin of a phenomena and focus on utilizing data-science, machine learning techniques, which may utilize neural networks (although not covered in the course). The goal is to study a relevant topic, which applies the methods covered in this course.

The midterm report will show details for the derivation of governing equations, discretization, and stability analysis. The final report will briefly include points covered in midterm along with other points mentioned above, such as result analysis and findings.

PREREQUISITE

MATH 2260, MATH 3132 and MATH 3202.

MATH 2260 is essential to gain knowledge of solution of linear differential equations and systems. Students with equivalent knowledge in computational techniques and calculus may take the course if they have not completed 3132/3202, subject to permission of course instructor.

TEXT BOOK

There will be no fixed textbook, but recommended books are listed below. Here, '*' indicates the books were primarily considered in organizing lecture notes.

- *Arieh Iserles, A First Course in the Numerical Analysis of Differential Equations
- Adler, James and Sterck, Hans De and Maclachlan, Scott and Olson, Luke, Numerical partial differential equations.
- *Finite difference methods for ordinary differential and partial differential equations by Randall LeVeque.
- Spectral methods in Matlab by Lloyd Trefethen (available in the pdf format).
- K. W. Morton and D. F. Mayers, Numerical Solution of Partial Differential Equations, Cambridge,
- Lecture notes by J. Alam (available only during lectures of math 4162).
- Any other text book of your choice.

Plagiarism and other academic misconduct:

Any instance of plagiarism or cheating or any other form of academic misconduct will be treated according to Section 6.12 of the University Calendar. Students are advised to consult the relevant sections for full details.

Note that section 6.12.4 includes the following statements of **Academic offences**:

- **Impersonating** another student or allowing oneself to be impersonated: This includes the imitation of a student or the entrance into an arrangement with another person to be impersonated for the purposes of taking examinations or tests or carrying out laboratory or other assignments.
- **Plagiarism**: Plagiarism is the act of presenting the ideas or works of another as one's own. This applies to all material such as essays, laboratory assignments, laboratory reports, work term reports, design projects, seminar presentations, statistical data, computer programs, research results and theses. The properly acknowledged use of sources is an accepted and important part of scholarship. Use of such material without acknowledgement is contrary to accepted norms of academic behaviour. Information regarding acceptable writing practices is available through the Writing Centre at www.mun.ca/writingcentre